1. What is data cleaning, and why is it important in data analysis?
   - What are the potential consequences of analyzing unclean or messy data?
   - Explain the common steps involved in cleaning and organizing data.

   Answer-

   Data cleaning, also known as data cleansing, is the process of finding and fixing (or removing) errors, mistakes, and messy parts in your data so that it becomes accurate, consistent, and ready for analysis.

   Data cleaning is one of the most important steps in data analysis because:

- Garbage In, Garbage Out (GIGO): If your data is wrong or messy, your results will also be wrong, no matter how smart your analysis tools or techniques are.
- It saves a lot of time later. Clean data makes analysis faster and more reliable.
- Good decisions can only be made from good quality data. Companies, researchers, and governments rely on clean data for important choices.
- Wrong conclusions: You might think sales are increasing when they are actually falling, or that a medicine is effective when it is not.
- Bad business decisions: A company may launch a product in the wrong city or waste money on wrong marketing because of incorrect customer data.
- Wasted time and money: You may have to repeat the entire analysis after discovering mistakes later.
- Loss of reputation: If your report or research is based on dirty data and someone finds out, people will stop trusting your work.
- Missed opportunities: Important patterns or problems in the data may remain hidden.

Common Steps Involved in Cleaning and Organizing Data

Here are the usual steps people follow while cleaning data (explained simply):

1. Understand the Data First, look at your data carefully. Check how many rows and columns there are, what each column means, and what kind of information it contains (numbers, text, dates, etc.).
2. Remove Duplicate Records Delete any rows that are exactly the same. For example, if the same customer's information appears twice, keep only one.
3. Handle Missing Values Find empty or blank cells. You can either:
   - Delete those rows (if not too many),
   - Fill them with average value, most common value, or a logical guess,

- Or leave them as “missing” and handle them during analysis.

4. Fix Incorrect or Inconsistent Data
    - Correct spelling mistakes (e.g., “New York”, “new york”, “NY” → make all the same).
    - Fix wrong formats (e.g., dates written as 12/01/2025 and 01-12-25 → make them uniform).
    - Convert text to numbers or numbers to text wherever needed.
5. Handle Outliers Find values that look strange or impossible (e.g., a person’s age as 150 years or salary as ₹1 crore for a school teacher). Decide whether to keep, correct, or remove them.
6. Standardize the Data Make everything consistent – same units (kg or pounds), same capital/small letters, same date format, etc.
7. Organize and Structure the Data Arrange columns in a logical order, give clear column names, and sometimes reshape the data (e.g., from wide format to long format) so it becomes easy to analyze.
8. Validate the Cleaned Data Finally, do a quick check – run some basic summaries, look at sample rows, and make sure the data now looks clean and makes sense.

Data cleaning usually takes the most time in any data analysis project (often 60-80% of the total time), but it is worth it because it makes the actual analysis part much easier and more enjoyable.

2. How would you sort the following dataset first by "Department" (A-Z) and then by "Salary" (Largest to Smallest)? Write a step-by-step approach.

EMPLOYEE	**DEPARTMENT**	**SALARY**
Sonu	IT	4000
Pranav	HR	5000
Rahul	IT	2500

Answer -

EMPLOYEE	DEPARTMENT	SALARY
Pranav	HR	5000
Sonu	IT	4000
Rahul	IT	2500

Steps -

1. Open your dataset in Excel Make sure your data is in a tabular format with clear column headers like:

- Employee
- Department
- Salary

Select the entire dataset Click on any single cell inside your data table. Excel will automatically detect the full range of your data. (Tip: If your data has filters or blank rows, it's safer to manually select the whole range including headers.)

Go to the Data tab In the ribbon at the top, click on the Data tab.

Click on the Sort button In the "Sort & Filter" group, click Sort. A "Sort" dialog box will appear.

Set up the first sorting level (Department)

- In the first row of the dialog box:
  - Sort by: Select Department
  - Sort On: Select Cell Values (this is the default)
  - Order: Select A to Z

Add the second sorting level (Salary)

- Click on the button Add Level (at the top of the dialog box).
- A new row will appear:
  - Then by: Select Salary
  - Sort On: Select Cell Values
  - Order: Select Largest to Smallest (this is descending order)

3. Explain the use of text functions such as TRIM , LEFT, RIGHT, MID, and CONCAT in data cleaning.
   Answer-

Use of Text Functions in Data Cleaning

Data cleaning means fixing messy text data so it becomes neat and useful. In Excel or Google Sheets, we use simple text functions like **TRIM**, **LEFT**, **RIGHT**, **MID**, and **CONCAT** to clean the data easily.

Here's what each function does:

1. TRIM This function removes extra spaces from the beginning, end, and between words. **Why we use it**: Sometimes names or addresses have unnecessary spaces, which can create problems later. **Example**: If the cell has " Hozefa Memon ", TRIM will make it "Hozefa Memon".

2. LEFT It takes characters from the left side of the text. **Example**: From "Hozefa Memon", LEFT can take the first name "Hozefa".

3. RIGHT It takes characters from the right side of the text. **Example**: From "9876543210", RIGHT can take the last 4 digits "3210".

4. MID This function takes text from the middle of a sentence or code. **Example**: From vehicle number "GJ01AB1234", MID can pull out "01AB" from the middle.

5. CONCAT It joins two or more pieces of text together. **Example**: It can combine "Hozefa" and "Memon" to make "Hozefa Memon". (You can also use & symbol instead of CONCAT)

Why are these functions important?

These functions help us:

- Remove unwanted spaces
- Split long text into useful parts
- Join short pieces into proper information
- Make the data clean, consistent, and ready for analysis

In short, TRIM, LEFT, RIGHT, MID, and CONCAT are simple but powerful tools that save a lot of time while cleaning messy data.

4. What is the role of date functions like TODAY in managing datasets?
   Answer -
   In data management, date functions help us work with dates easily and keep the data up-to-date. One of the most useful functions is **TODAY**.

What is TODAY function? The TODAY function automatically shows the **current date** in the cell. **Formula**: =TODAY()

Role in Managing Datasets:

- **Keeps data fresh**: It always updates to today's date whenever you open the file. No need to change the date manually every day.
- **Calculates age or days**: You can easily find out how many days have passed since a certain date. Example: To calculate a person's age in days → =TODAY() - A2 (where A2 has the birth date)
- **Tracks deadlines**: It helps check whether a task is overdue or still on time. Example: =TODAY() - B2 (if result is positive, the deadline has passed)
- **Adds current date automatically**: Useful when you want to record when a new entry was added in a dataset (like customer registration date).
- **Helps in reports**: Many reports need "Date of Report" or "As on Date". TODAY puts the latest date without any extra work

5. Apply Data Validation to restrict Quantity values to only whole numbers between 1 and 10.
   a. Configure an input message that appears when a user selects a cell in the "Quantity" column explaining: "Please enter a whole number between 1 and 10."
   b. Set up an error alert message that triggers if the user enters a number less than 1 or greater than 10, showing: "Invalid input! The quantity must be a whole number between 1 and 10."

Customer Name	Product Name	Category	Quantity	Unit Price($)
Jane Smith	Shoes	Electronics		81
Isabella Moore	Laptop	Electronics		121
Daniel Davis	Sofa	Clothing		239
Alex Moore	Shoes	Electronics		500
Michael Johnson	Table Lamp	Home Décor		423
Daniel Johnson	Backpack	Electronics		160
Isabella Davis	Headphones	Electronics		348
Jane Davis	Headphones	Electronics		152
Alex Wilson	T-shirt	Home Décor		369

Answer -
Data set